

4 Goodness of fit and contingency tables

What students need to learn:

The null and alternative hypotheses. The use of $\sum_{i=1}^n \frac{(O_i - E_i)^2}{E_i}$ as an approximate χ^2 statistic.

Degrees of freedom.

Applications to include the discrete uniform, binomial, Normal, Poisson and continuous uniform (rectangular) distributions. Lengthy calculations will not be required.

Students will be expected to determine the degrees of freedom when one or more parameters are estimated from the data. Cells should be combined when $E_i < 5$. Yates' correction is not required.

5 Regression and correlation

What students need to learn:

Spearman's rank correlation coefficient, its use, interpretation and limitations.

Testing the hypothesis that a correlation is zero.

Numerical questions involving ties will not be set. Some understanding of how to deal with ties will be expected.

Use of tables for Spearman's and product moment correlation coefficients.

Unit S4 Statistics 4

GCE Further Mathematics and GCE AS and GCE Further Mathematics (Additional)
A2 optional unit

S4.1 Unit description

Quality of tests and estimators; one-sample procedures;
two-sample procedures.

S4.2 Assessment information

Prerequisites

A knowledge of the specification for S1, S2 and S3 and its prerequisites and associated formulae is assumed and may be tested.

Examination

The examination will consist of one 1½ hour paper. The paper will contain about seven questions with varying mark allocations per question which will be stated on the paper. All questions may be attempted.

Calculators

Students are expected to have available a calculator with at least the following keys: $+$, $-$, $\times$, $\div$, π , x^2 , $\sqrt{x}$, $\frac{1}{x}$, x^y , $\ln x$, e^x , $x!$, sine, cosine and tangent and their inverses in degrees and decimals of a degree, and in radians; memory. Calculators with a facility for symbolic algebra, differentiation and/or integration are not permitted.

Formulae

Students are expected to know any other formulae which might be required by the specification and which are not included in the booklet, *Mathematical Formulae including Statistical Formulae and Tables*, which will be provided for use with the paper. Questions will be set in SI units and other units in common usage.

Students will be expected to know and be able to recall and use the following formulae:

Type I error: $P(\text{reject } H_0 \mid H_0 \text{ true})$

Type II error: $P(\text{do not reject } H_0 \mid H_0 \text{ false})$